CHE301 current papers for midterm spring 2018

1) electrophoresis _____ method may be used to separate samples using chromatography

2) **The life line of a heart patient as appeared in ECG is random movement the linearity means?** _____Death
3) how many QC tools are _____7____.
4) __quality control___ is a part of quality assurance.
5) __scatter diagram___ can identify cause and effect relation.
6) RSD canot __be negative____ while coefficient of variation
7) A __standard___error is the standard deviation.
8) number of significance figure 9.00823 are?
9) standar deviation is square root of _____verience
10) Calculation of error relative to actual value is called realtvie error
11) **Indirect Titration** or alkalimetrictitration ____is used for determination of ions such as anions which do not react with EDTA chelon.
12) **Moderent black II at pH 6-7 changes color red to blue**

13) Error free data is called? ____Accurate
14) Application of the analytical procedure to synthetic mixtures of the drug product components to which known quantities of the drug substance to be analyzed have been added it is called____ Assay of Drug product
15) **Sublimation**To separate solid fromsolids,
16) LOQ is a std dev
17) Second type of error is determinant error
18) Definition of rainacidity____
19) Quality record is not includes_____
20) LOQ pH ratio_____10:1
21) The sample from which we separate these analytes is called matrix
22) In aspects of reportingresults Reporting of data- includes Standard error of mean, standard deviation of the sampling distribution, Coefficient of variance all
23) smallest quantity of analyte that can be detected_____ Limit of Detection
24) Particle are visible in the naked eyes10^{-7} - 10^{-4} cm in diameter lowest concentration of analyte in a sample with acceptable accuracy_ Limit of Quantitation ____.
25) Methyle blue at PH 4-5 changes color_____blue to yellow__
26) Spectroscopy is used for _____light detection

27) What is mathematical expression of beer Lambert law? (2)

I0 = Incident light, I = Transmitted light, l = path length, C = Concentration, · = Molar Absorptivity

28) Name two test hypotheses?

Inferential > 1 Variable > 2 Samples > T-Test

T test:

$$t = \frac{\overline{X} - \mu}{\frac{s}{\sqrt{N}}}$$

F test = Variance between/variance within

29) Name the test for varience?

Analysis of Variance- Z-test, T-test, F-Test

30) Write the procedure for draw a histogram?

When creating a histogram, "a range of data is divided into smaller sections having a uniform span and the number of data contained in each section (the number of occurrences) is counted to develop a frequency distribution table." Then, "a graph is formed from this table by using vertical bars, each having the height proportional to the number of occurrences in each section." It is Used to analyze a process to identify a problem point that needs to be improved by finding the location of the mean value or degree of variations in the graph.

31) Write detailed uses of NMR spectroscopy and mass spectroscopy.

NMR Spectroscopy- Used for structure for quantitation elucidation and not

Mass Spectrometry- Used as detector with both GC MS and ICP MS

32) WhAT is the standerd daviation used for?

Std Dev is measure of precision. It shows how close is your data in a data set. It denote with sigma.

33) What do you know about pareto diagram?

Pareto Diagram is a combination of bar and line graphs of accumulated data, It is used where data associated with a problem e.g. a defect, or mechanical failure, or a complaint from a customer are divided into smaller groups by cause or by phenomenon and sorted by the number of occurrences or the amount of money involved. (The name "Pareto" came from an Italian mathematician who created the diagram.) Bar chart arranged in descending order of height from left to right, bars on left relatively more important than those on right. Separates the

"vital few" from the "trivial many" (Pareto Principle). **Benefits-** Breaks big problem into smaller pieces, Identifies most significant factors, Shows where to focus efforts, Allows better use of limited resources.

34) Name the sources of errors?

Sources of error- Instrumental error, Method error, Human error. There are just three main factors that can be named as sources of error. **Instrumental Error-** If a balance weigh 2 g as 2.1 g. If a 10 mL pipette draw a volume of 9.5 mL on average. If a faulty pH meter read 4.0 pH buffer as 4.21 even after calibration. Leakage in burette The **second er**ror type is method error. If your analytical method have some limitations then itmay generate error. Wehave to segregate method error fromhuman error. **Method Error-** If a method say, heat it, stay at room temperature, use buffer of pH 10. Note: The meaning ofsuch statements may be different for different analyst. Third type of errors are personnel error. Analyst error Human Error (Personnel error)- If a personnel- Problem in identifying colour / meniscus, Don't follow procedure properly, Careless behavior in calibration, Note the meniscus above or below eye level. Like if he watch the meniscus above or beloweye level. Or fromfilling of pipette to pouring of sample it drop the sample

35) Goals of sampling.

Goals of sampling: SampleSize, Sampling types, Sampling techniques, Samplepreparations, To obtain a mean analyte concentration, Unbiased estimate of the population mean. Every possible sample is equally and likely to be drawn

36) What is limit of quantitation?

Limit of Quantitation (LOQ): It is the lowest concentration of an analyte in a sample that may be determined with acceptable accuracy and precision.

37) How equipment can be calibrated?

Calibration of an analytical balance using standard weights, Calibration of pipette / micropipettes and other volumetric apparatus to SI units using a calibrated balance, Calibration of pH meter with standard buffers.

38) Give the sequence of quality assurance from SI units to products.

Quality Assurance_____> Quality Manager, R&D, NPD, QualityControl

39) Define signal to noise ratio?

Signal to Noise Ratio (3 or 2:1 is acceptable of LOD while 10:1 is generally acceptable for LOQ).

40) Find the relative error?

Calculation of error relativeto actual value is called relativeerror. rror may rise with sample volume, but relativeerror will remain same Problem- If 0.5mg of precipitate is lost as a result of washing and the precipitate actually weighs 500mg , what will be the relative error due to solubility. It is calculated in percentage. Or part per thousands PPT. 0.5/5000*100= 0.1 % = 1 PPT. Similarly in pipette if instead of 10 mLit shows 9.5 mLaverage or an error of 0.5 mL. So relative error may calculate 0.5/10*100 =5%. We can compare one pipette from other in accuracy. Or we can compare and declare that a burette is more reliable than a pipette.

41) Define relative standard deviation?

relative standard deviation (RSD) is a special form of the Std Dev that is always positive. The **relative standard deviation** (RSD) is a special form of the **standard deviation** (std dev). It's generally reported to two decimal places (i.e. an RSD of 2.9587878 becomes 2.96). As the denominator is the absolute value of the mean, the RSD will always be positive **FormulaRSD: s/x.100%** However, the RSD cannot be negati

42) Main types of error?

Error and types of error- Systematic error, Random error

43) What is error?

Error simply means not correct. But mathematically itmay have some other meanings as well Absolute Error (E) · X1 · X2

44) Write down the Types of sample?

Sampling types- there are three types of sample

Grab sample: Sample may collect at some particular time and location are called grab sample

Composite Samples: many grab samples may combine to form composites

Gross samples: gross sample may be huge is determined by the uncertainty that can be tolerated between the composition of the gross sample, the degree of heterogeneity, the level of particle size at which heterogeneity begins.

45) What factors needed for instrumental techniques

HPLC (high performance liquidchromatography

GC (Gas chromatography

46)classification of gravimetric analysis

a set of method used in analytical chemistry for the quantitative determination of an analyte based on its mass there are four main types of it.

Precipitation gravimetry, the analyte is separated from a solution of the sample as a precipitate and is converted to a compound of known composition that can be weighed.

Volatilization gravimetry, the analyte is separated from other constituents of a sample by conversion to a gas of known chemical composition. The weight of this gas then serves as a measure of the analyte concentration.

Electrogravimetry, the analyte is separated by deposition on an electrode by an electrical current. The mass of this product then provides a measure of the analyte concentration.

47) Write steps of gravimetric analysis?3

There are some seteps

Steps required in gravimetric analysis:

- **Preparation of the solution**
- **Precipitation**
- **Digestion**
- **Washing**
- **Drying or igniting**
- **Weighing**
- **Calculation**

medical students (bio_ technologist)

48)Colloidal suspension crystalline suspension

Types of precipitates (on basis of particle size)

1-Colloidal suspensions

- Particles are invisible to the naked eye (10^{-7} - 10^{-4} cm in diameter).
- Colloidal particles show no tendency to settle from solution
- not easily filtered.

2-Crystalline suspension

- particles with dimensions on the order of tenths of a millimeter or greater.
- The temporary dispersion of such particles of a tend to settle spontaneously,
- easily filtered.

49) Write the main procedures that can be used in analytical chemistry?5

Wet analysis, instrumental analysis, Instrument analysis is the calibration of apparatus and equipment, SOPs, STMs, IQ/OQ/PQ of equipment •Evaluation of analytical data
Sample and Population, Significant figures, Mean, Mode, Median, UoM, Std.Dev., Variance etc) Reporting of results We may use an instrumental method of analysis or wet
analysis. Certain wet analysis methods are acid base titration, redox titration, complexometric titration or gravemetric analysismethods. If we choose instrumental method, the instrument should be calibrated. Should follow proper standard operating procedure and standard testing methods. Instrument should be installed, operated and performed according to standard specifications. And the last but not least is statistical evaluation of analytical data. Wewill learn in future modules about sample and population of analytical data. Significance of significant figures. Calculation of mean, median and mode of available data and measurement of uncertainty, standard deviation and variance etc. Finally we report our analytical results along with standard specifications, using proper units of measurements and all related informations or logos on report.

50) How errors are removed

Removal of error-Calibration of Equipment,

Validation of analytical method. Training of personnel. We have

discussed three possible causes of error. Equipment, method

and personnel. So three possible solutions. Equipment can be

used properly after calibration. Calibration of Equipment-

Calibration of an analytical balance using standard weights,

Calibration of pipette / micropipettes and other volumetric

apparatus to SI units using a calibrated balance, Calibration of

pH meter with standard buffers.

51)Which factor we consider in prior to method validation?

Validation of analytical methods- Analytical procedure is suitable for its intended purpose. Validate all non- standard analytical methods. Validated by R&D before being transferred to the QC . Quality Assurance- Considerations Prior to MethodValidation

Suitability of Instrument- Status of Qualification and Calibration

Suitability of Materials- the Status of Reference Standards, Reagents, etc.

Suitability of Analyst- Status of Training and Qualification Records

Suitability of Documentation- Written analytical procedure and proper approved protocol with pre-established acceptance criteria.

52)Goals for sampling

Goals of sampling

SampleSize, Sampling types, Sampling techniques, Samplepreparations, To obtain a mean analyte concentration, Unbiased estimate of the population mean. Every possible sample is equally and likely to be drawn. **Gross Sample Size** is determined by the uncertainty that can be tolerated between the composition of the gross sample, the degree of heterogeneity, the level of particle size at which heterogeneity begins.

53) Limit of detection

…Limit of Detection (LOD): It is the smallest quantity of an analyte that can be detected, and not necessarily determined in a quantitative fashion.

54) Define analyte?2

An **analyte**, **component** (in clinical chemistry) or **chemical species**, is a substance or chemical constituent that is of interest in an analytical procedure. Whilein chemical analysis,

55) Two examples of scatter diagram?2

Relationship between thermal treatment temperature of a steel material and its tensile strengths, Relationship between visit made by a salesman and volume of sales

56) Coefficient of variation?3

relative standard deviation (coefficient of variation), The **coefficient of variation** is a measure of spread that describes the amount of **variability** relative to the mean. Because the **coefficient of variation** is unitless, you can use it instead of the standard deviation to compare the spread of data sets that have different units or different means. Fprmula: S/X.100%

57) Detail cause and effect diagram?5

Cause & Effect Diagram- Also called fish- bone or Ishikawa

diagram presents a systematic representation of the relationship between the effect (result) and affecting factors (causes)." It is also called fish bone diagram, or Ishikawa diagramis a tool to identify the root cause that is responsible for an effect. Solving a problem requires clarification of a cause and effect relationship, where the effect (e.g., the result of work) varies according to factors (e.g., facilities, machines used, method of work, workers, materials and parts used etc.). To obtain a good work result, we must identify the effects of various factors and developmeasures to improvethe result accordingly.

Can obtain a clear overall picture of causal relation. (A change in the cause triggers a variation in the result.)

Can list up all causes to identify important causes. Can determine the direction of action (counter measure).

58) Two types of chromatography?(2)

59) What is linearity?(3)

60) What is presipitation gravimetry?(3)